Sequence and State Diagrams

Smart Emergency Response & Healthcare Coordination Platform

1. Overview

This document shows how the system parts talk to each other during the main scenarios (**Sequence Diagrams**) and all the statuses an emergency request goes through (**State Diagram**).

The diagrams follow the **Use Cases (`UC-01` to `UC-21`)**, **User Stories (`US-PAT`, `US-OPR`, `US-HOS`, `US-AMB`, `US-DOC`)**, **Functional Requirements (`FR-10` to `FR-40`)**, **Non-Functional Requirements (`NFR-PERF`, `NFR-REL`, `NFR-DAT`)** and the **System Architecture**.

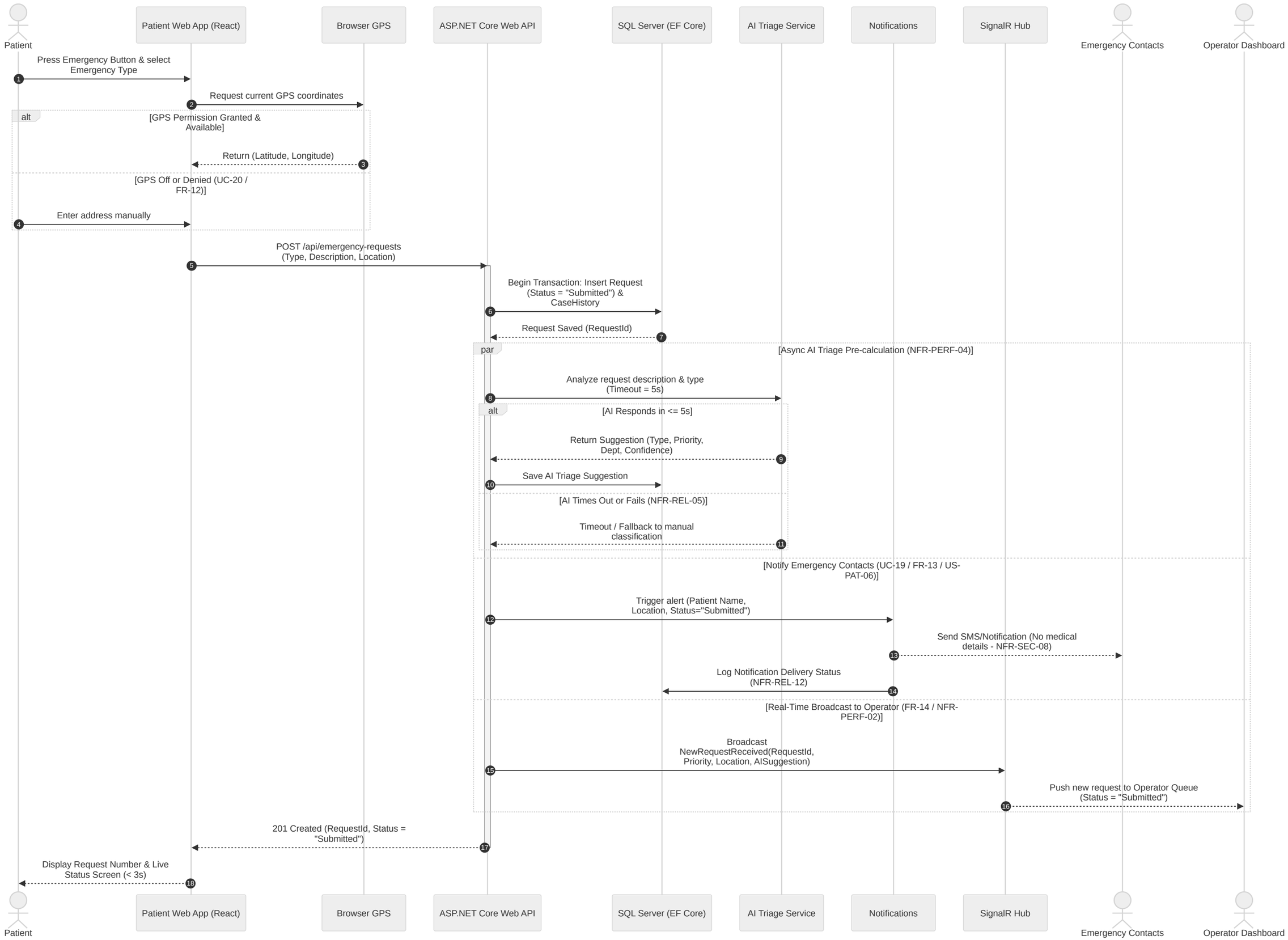
Main System Participants & Components

- **Actors:** Patient, Operator, Hospital Staff, Paramedic, Doctor, Emergency Contacts.
- **Frontend Clients:** React Web Apps (Patient App, Paramedic App, Operator/Hospital/Doctor Dashboards).
- **Backend:** ASP.NET Core Web API (Emergency Requests, Hospital Matching, Ambulance Matching, Cases & Status, Notifications).
- **Real-time:** SignalR Hub.
- **AI Triage Service:** suggests the type and priority only.
- **Other:** SQL Server Database (EF Core), Google Maps API, the browser GPS.

2. Sequence Diagrams

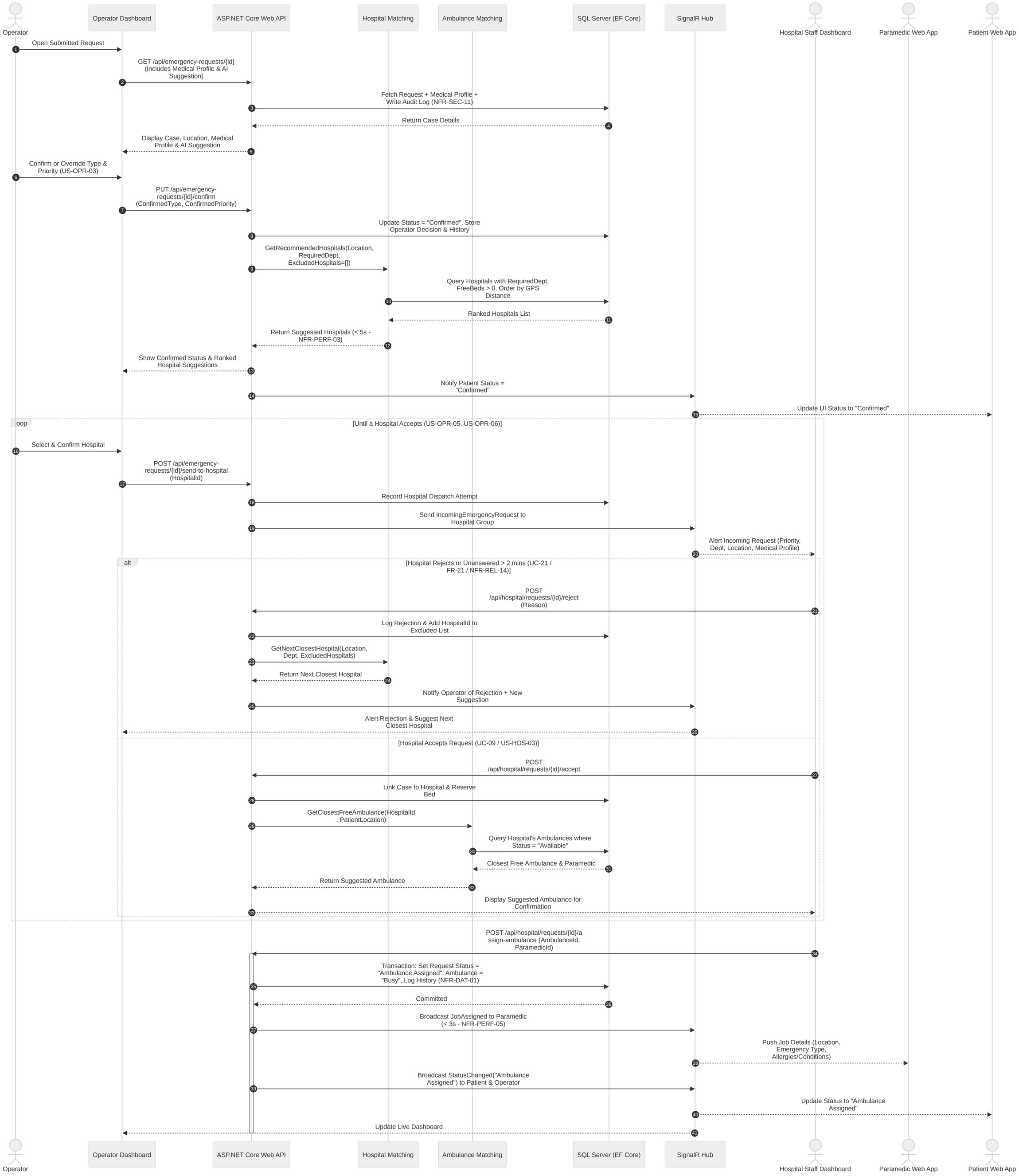
2.1. Scenario 1: Emergency Request Submission (`UC-04`, `UC-19`, `UC-20`, `FR-10` to `FR-14`, `US-PAT-01`)

The patient presses the emergency button. The app takes the GPS location (or he writes his address if GPS is off), the request is saved first (`NFR-REL-03`), the AI gives its suggestion, the emergency contacts get a message (`NFR-REL-12`) and the request shows on the Operator dashboard in less than 3 seconds (`NFR-PERF-01`, `NFR-PERF-02`).



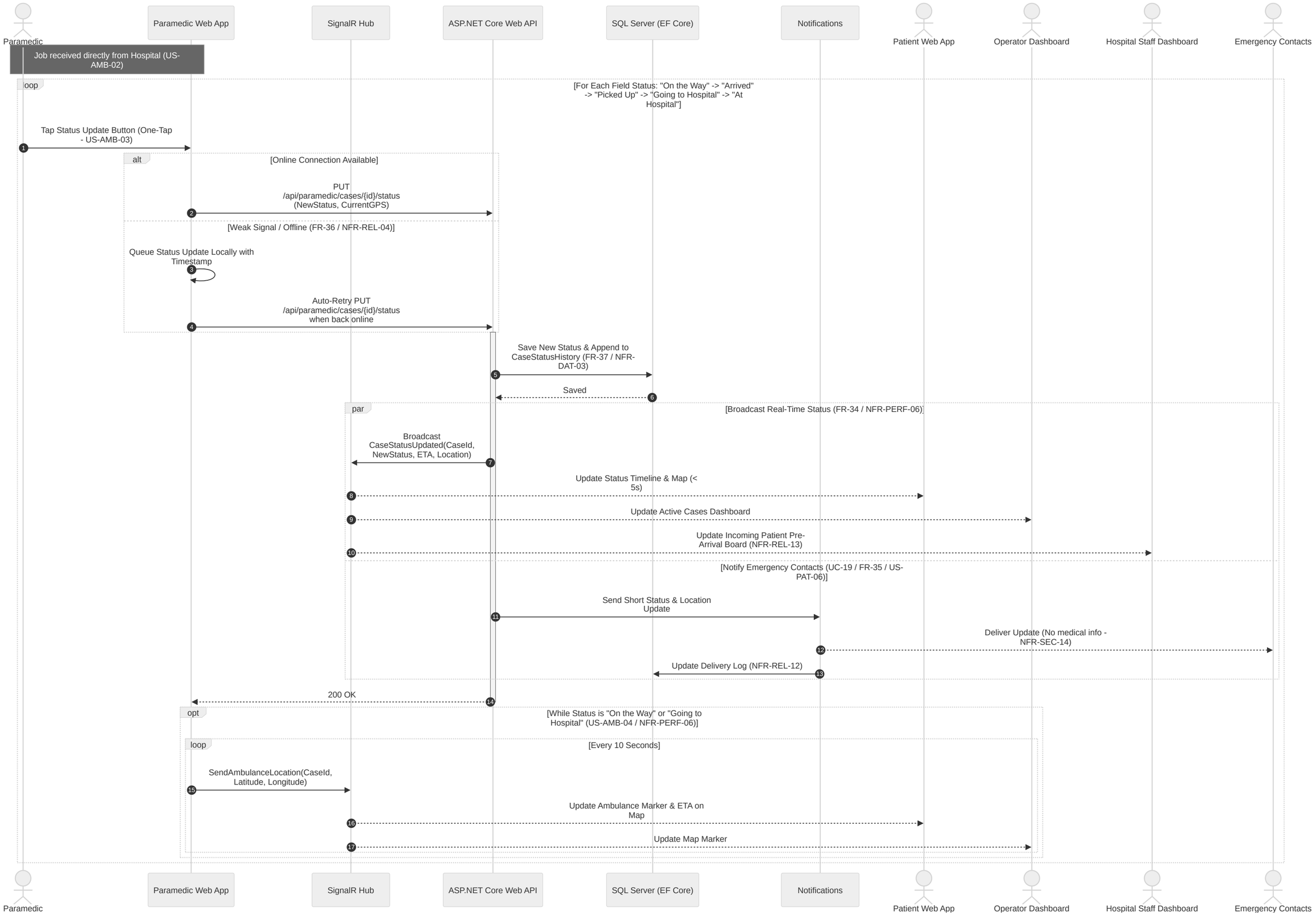
2.2. Scenario 2: Request Review, Hospital Selection & Ambulance Assignment (`UC-06`, `UC-07`, `UC-09`, `UC-21`, `FR-17` to `FR-27`)

The Operator reviews the AI suggestion and confirms it (`US-OPR-02`, `US-OPR-03`), then sends the request to the suggested hospital. If the hospital rejects, the next closest hospital is suggested (`US-OPR-05`, `US-OPR-06`). When a hospital accepts (`US-HOS-03`), it sends its closest free ambulance with a paramedic (`US-HOS-04`, `US-AMB-02`).



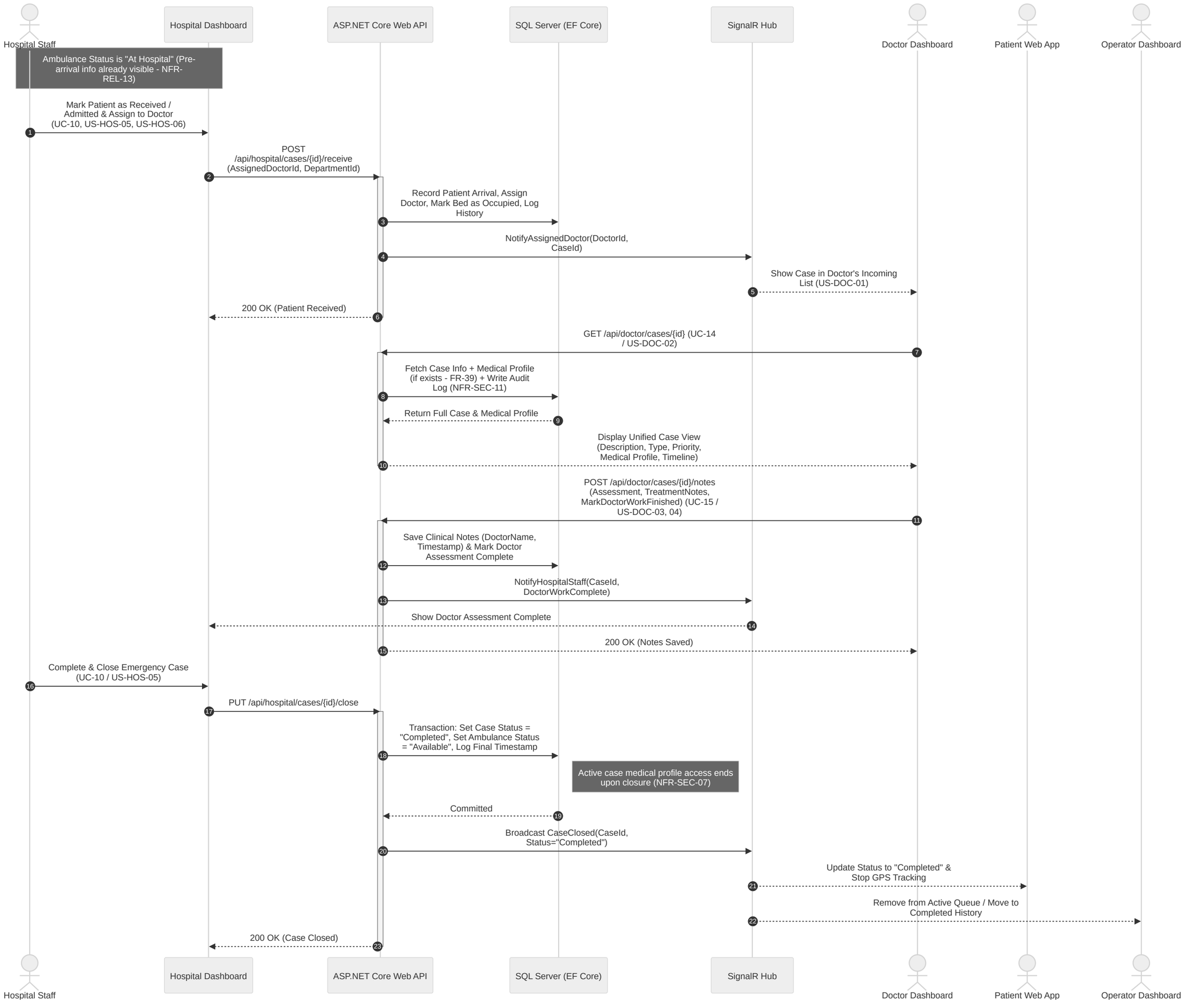
2.3. Scenario 3: Live Location Tracking & Status Updates (`UC-05`, `UC-12`, `UC-13`, `FR-15`, `FR-16`, `FR-31 to FR-37`)

The paramedic updates the status with one tap (`US-AMB-03`) and his location is sent every 10 seconds (`US-AMB-04`, `NFR-PERF-06`). If the signal is weak, the update is sent when he is back online (`FR-36`). The emergency contacts get a short update with no medical info (`UC-19`, `FR-35`).



2.4. Scenario 4: Hospital Reception, Doctor Assessment & Case Closure (`UC-10`, `UC-14`, `UC-15`, `FR-28` to `FR-30`, `FR-38` to `FR-40`)

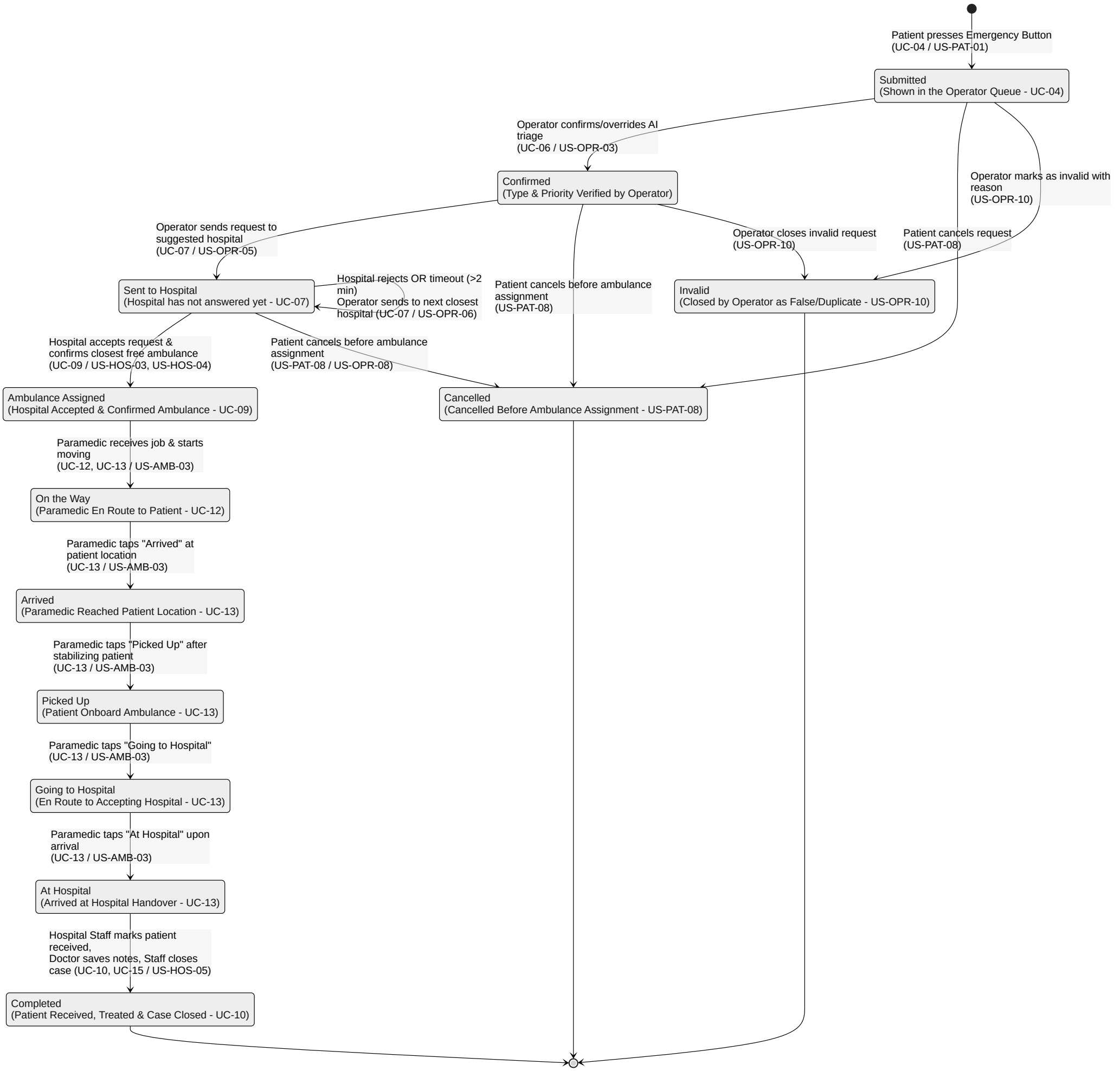
When the ambulance reaches the hospital (`At Hospital`), the staff receive the patient and give the case to a doctor (`US-HOS-06`). The doctor sees the case and writes his notes (`US-DOC-01` to `04`). Then the staff close the case (`US-HOS-05`), the ambulance becomes free again and access to the medical profile ends (`NFR-SEC-07`).



3. Emergency Request State Diagram

This diagram shows all the statuses of an **Emergency Request**. The main statuses are the ones in **US-PAT-02**. We also added **Sent to Hospital** (the hospital has not answered yet, UC-07), **Cancelled** (US-PAT-08) and **Invalid** (US-OPR-10). These three must also be in the status list of the Database Schema.

3.1. State Diagram (Mermaid stateDiagram-v2)



3.2. State Transitions & Consistency Reference Table

The table shows every status change, who does it and the rule behind it.

#	Current State	Event / Trigger	Next State	Triggered By	Guard Condition / Business Rule	Related Requirement
1	*(Initial)*	Submit Emergency Request	`Submitted`	Patient	GPS captured or manual address entered; saved in DB first.	`UC-04`, `US-PAT-01`, `FR-14`
2	`Submitted`	Review & Confirm Priority	`Confirmed`	Operator	Operator confirms or overrides AI triage suggestion.	`UC-06`, `US-OPR-03`, `FR-18`
3	`Submitted` / `Confirmed` / `Sent to Hospital`	Cancel Request	`Cancelled`	Patient / Operator	Allowed **only** before status reaches `Ambulance Assigned`.	`US-PAT-08`, `US-OPR-08`
4	`Submitted` / `Confirmed`	Mark as Invalid	`Invalid`	Operator	Reason must be recorded (false or duplicate request).	`US-OPR-10`
5	`Confirmed`	Send Request to Hospital	`Sent to Hospital`	Operator	Hospital has required department and free beds.	`UC-07`, `US-OPR-05`, `FR-20`
6	`Sent to Hospital`	Reject Request / Timeout	`Sent to Hospital`	Hospital Staff / System	Rejecting hospital is excluded; next closest GPS hospital suggested.	`UC-07`, `UC-21`, `US-OPR-06`, `FR-21`, `NFR-REL-14`
7	`Sent to Hospital`	Accept & Confirm Ambulance	`Ambulance Assigned`	Hospital Staff	Hospital has free bed + free paramedic; ambulance marked `Busy`.	`UC-09`, `US-HOS-03`, `US-HOS-04`, `FR-27`
8	`Ambulance Assigned`	Update Status to En Route	`On the Way`	Paramedic	Paramedic taps status button; live GPS sharing starts.	`UC-12`, `UC-13`, `US-AMB-03`, `FR-33`
9	`On the Way`	Update Status at Scene	`Arrived`	Paramedic	Paramedic reaches patient coordinates; contacts notified.	`UC-13`, `US-AMB-03`, `FR-33`, `FR-35`
10	`Arrived`	Update Status on Boarding	`Picked Up`	Paramedic	Patient is onboard the ambulance.	`UC-13`, `US-AMB-03`, `FR-33`
11	`Picked Up`	Start Transit to Hospital	`Going to Hospital`	Paramedic	Ambulance navigates back to its home hospital.	`UC-13`, `US-AMB-01`, `US-AMB-03`
12	`Going to Hospital`	Arrive at Hospital	`At Hospital`	Paramedic	Ambulance arrives at hospital entrance.	`UC-13`, `US-AMB-03`, `FR-33`
13	`At Hospital`	Receive Patient & Close Case	`Completed`	Hospital Staff *(with Doctor notes)*	Patient admitted, Doctor notes saved, Ambulance freed (`Available`), temporary medical access revoked.	`UC-10`, `UC-15`, `US-HOS-05`, `NFR-SEC-07`